

Lab Report No 7
Understanding of ANN functions and learning rules
Artificial Intelligence



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Lab Task No 1:

Solution:

Brief description (3-5 lines)

ANN functions and learning rules:
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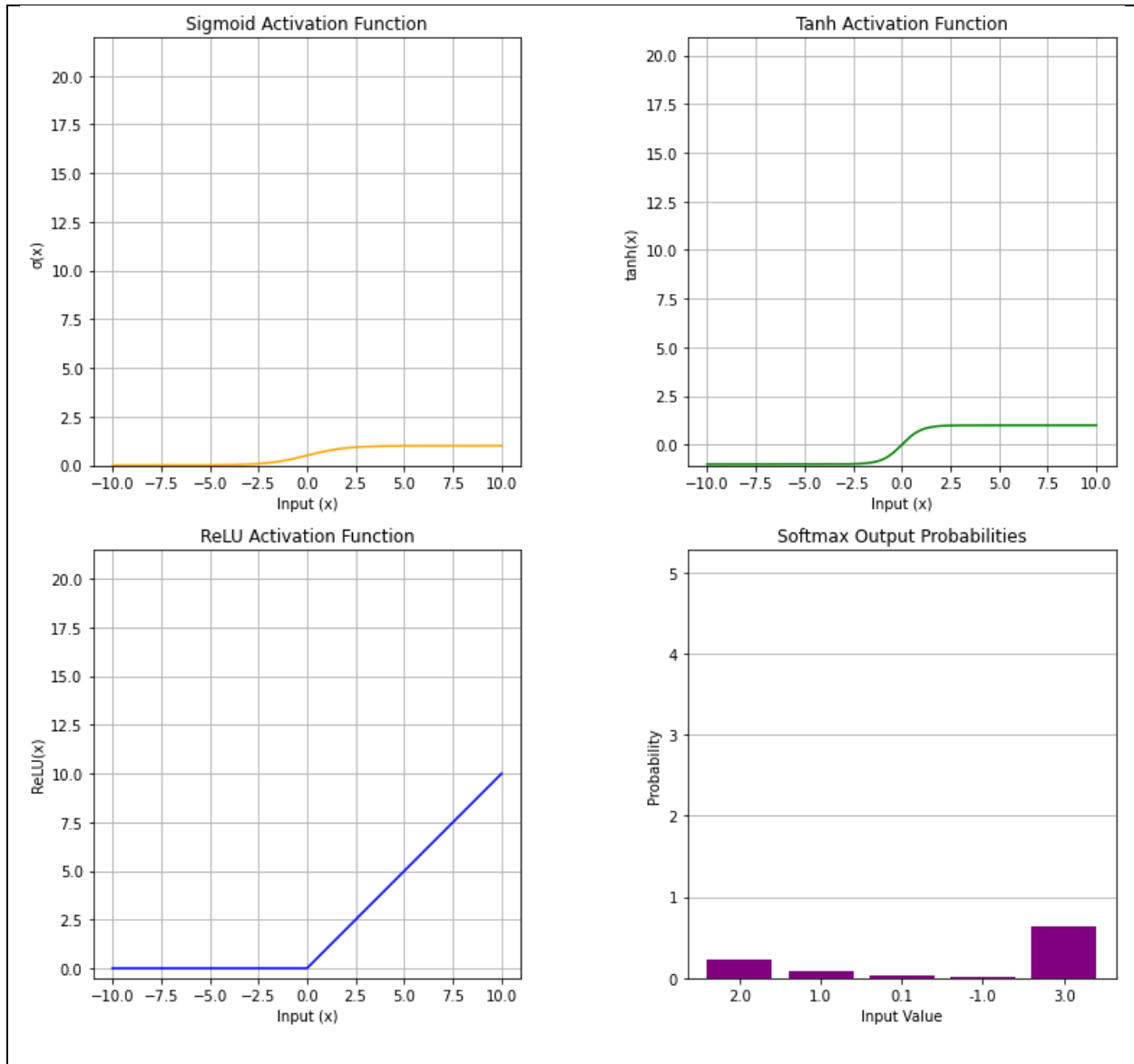
Write a user defined Python function to generate the following explained non-linear activation functions that are being used in neural networks using basic equations. Also plot them showing grid lines, title and x-label. Use axis square.

The code

<pre>import numpy as np import matplotlib.pyplot as plt # Activation Functions def sigmoid(x): return 1.0 / (1.0 + np.exp(-x)) def tanh(x): return np.tanh(x) def relu(x): return np.maximum(0, x) def softmax(z): e = np.exp(z - np.max(z)) return e / np.sum(e) x = np.linspace(-10, 10, 400) z = np.array([2.0, 1.0, 0.1, -1.0, 3.0]) softmax_probs = softmax(z) plt.figure(figsize=(12, 10)) plt.subplot(2, 2, 1) plt.plot(x, sigmoid(x), color='orange') plt.title("Sigmoid Activation Function") plt.xlabel("Input (x)") plt.ylabel("$\sigma(x)$") plt.grid(True) plt.axis('square') # Tanh Plot plt.subplot(2, 2, 2) plt.plot(x, tanh(x), color='green') plt.title("Tanh Activation Function") plt.xlabel("Input (x)") plt.ylabel("tanh(x)") plt.grid(True)</pre>

```
plt.axis('square')
# ReLU Plot
plt.subplot(2, 2, 3)
plt.plot(x, relu(x), color='blue')
plt.title("ReLU Activation Function")
plt.xlabel("Input (x)")
plt.ylabel("ReLU(x)")
plt.grid(True)
plt.axis('square')
plt.subplot(2, 2, 4)
labels = [str(val) for val in z]
plt.bar(range(len(z)), softmax_probs, color='purple')
plt.xticks(range(len(z)), labels)
plt.title("Softmax Output Probabilities")
plt.xlabel("Input Value")
plt.ylabel("Probability")
plt.grid(True, axis='y')
plt.axis('square')
plt.tight_layout()
plt.show()
```

The results (Screenshot)



Conclusion

In conclusion, we had implemented the artificial neural network activation graphs and also we implemented the different graphs of network activation functions such as Sigmoid, Tanh, ReLU, Leaky ReLU, Parametric ReLU, ELU, SELU, Swish, GELU and SOFTMAX PROBABILITY.